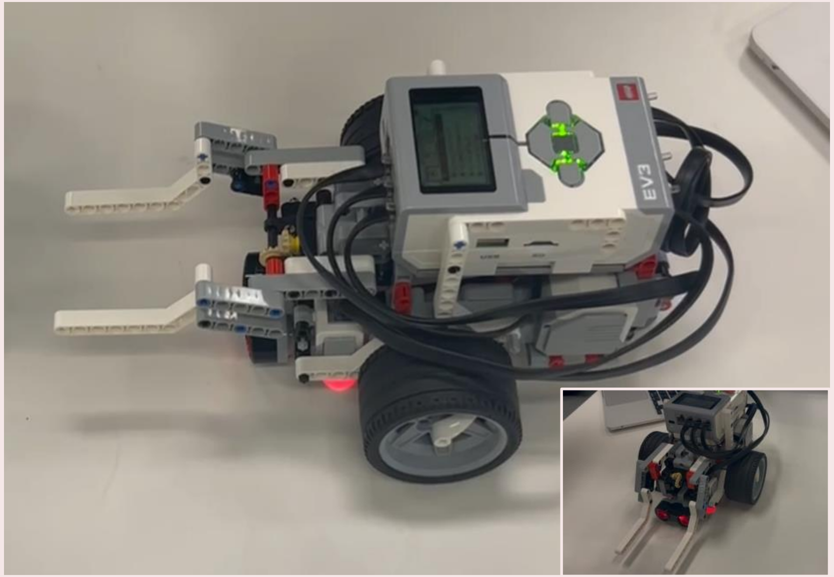
PROTOTYPE

Autonomous Sorting Robot

PMR Design Bid Proposal

ENED 1002c · Spring 2026 · University of Cincinnati

Submitted to: BC Engineering & Eco-Entrepreneurs



Team 10.15

Mahi Sheth · Kritika Dixit · Joy OlaOlorun

Meet The Team

Mahi Sheth

Project Manager

PRIMARY CONTRIBUTIONS:

Gantt chart & scheduling
Design notebook updates w/ meeting minutes + notes
Project coordination
Testing Plans
PowerPoint Presentation

Joy OlaOlorun

Lead Robotics Builder

PRIMARY CONTRIBUTIONS:

Robot physical construction
Robot arm design & fabrication
Hardware integration
Testing Plans

Kritika Dixit

Lead Programmer

PRIMARY CONTRIBUTIONS:

LabVIEW VI development
Sensor threshold calibration
SubTask 1 demo lead
Sorting logic coding

Global Context & Problem Statement

- ⚠ Natural disasters increasing in frequency & cost
- ♻ Debris = organic + metal waste — currently landfilled
- Thermal Depolymerization (TDP) converts debris → clean fuel (no net CO_2)
- Need: autonomous robot to sort & transport bins

TDP Process: Shred → Slurry → Heat & Pressure → Flash/Separate → Final Products | 80%+ efficient, zero emissions

Stakeholders & Social Impact

🏠 Displaced Residents

Faster debris removal

Cleaner community recovery

Clean energy from waste

🏛 Local Governments

Lower restoration costs

Energy microgrids from TDP

Less dependence on external aid

🌿 Environmental Organizations

Net-zero CO_2 process

Waste recycled, not landfilled

Zero hazardous byproducts

👷 Disaster Relief Workers

Robot handles hazardous transport

Reduces human risk exposure

Safer cleanup operations

⚡ Local Utility Companies

New local fuel source

Grid stabilization support

Reduced imported fuel reliance

Design Criteria & Specifications

Line Following
≥ 8/10 loops accurate
Solid & broken line types

Material Sorting
Organic vs. metal ≥ 90%
Detected via bin weight

Bin Transport
Correct station ≥ 3/5 tries
Organic: near · Metal: far

Auto Return
Return to shredder ≥ 90%
Follows closed loop back

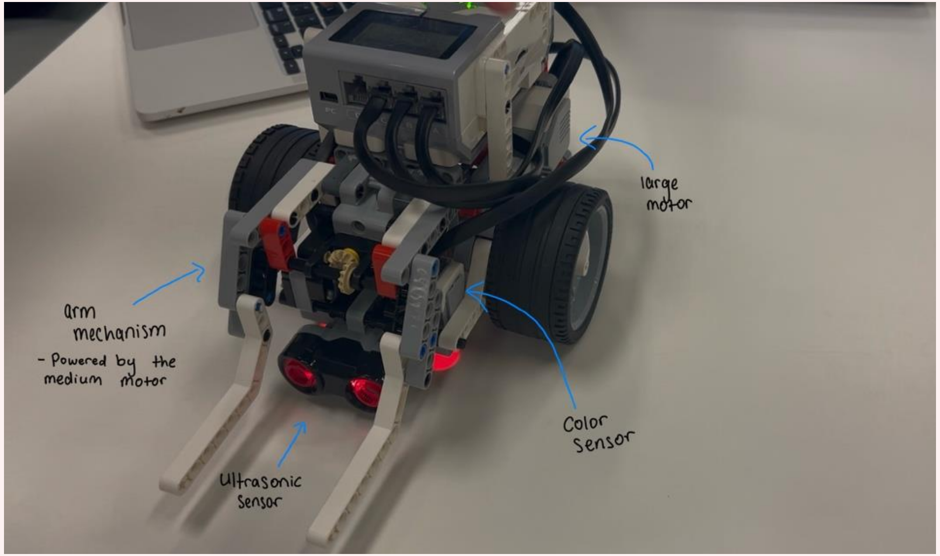
Container Exit
Shredder in ≤ 120 sec
Follows black ground line

Terrain Handling
Height variation ≤ 2 cm
No tipping allowed

Documentation
Block diagram · Gantt chart
Design notebook

Sustainability
LEGO kit parts only
Eco-Entrepreneurs values

Robot Design Overview



Locomotion

Two large drive wheels + rear caster
Handles up to 2 cm terrain variation

Navigation

2 downward-facing color sensors
Measures reflected light intensity

Bin Retrieval

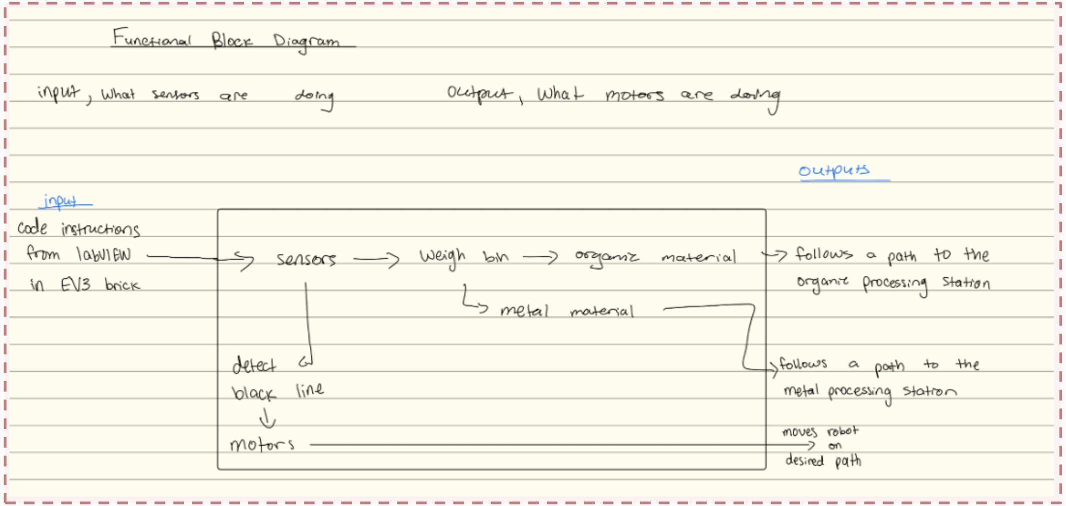
Mechanical arm + EV3 motor
Runs medium motor for 2 seconds

Contents Detection

Motor load while lifting ≈ bin weight
Classifies: light = organic · heavy = metal

LabVIEW: System Architecture & Flow

VI #1



INPUT	SENSORS	PROCESS	OUTPUT
LabVIEW code via EV3 Brick	Color sensors + Touch sensors	Weigh bin + Classify material	Navigate to correct station

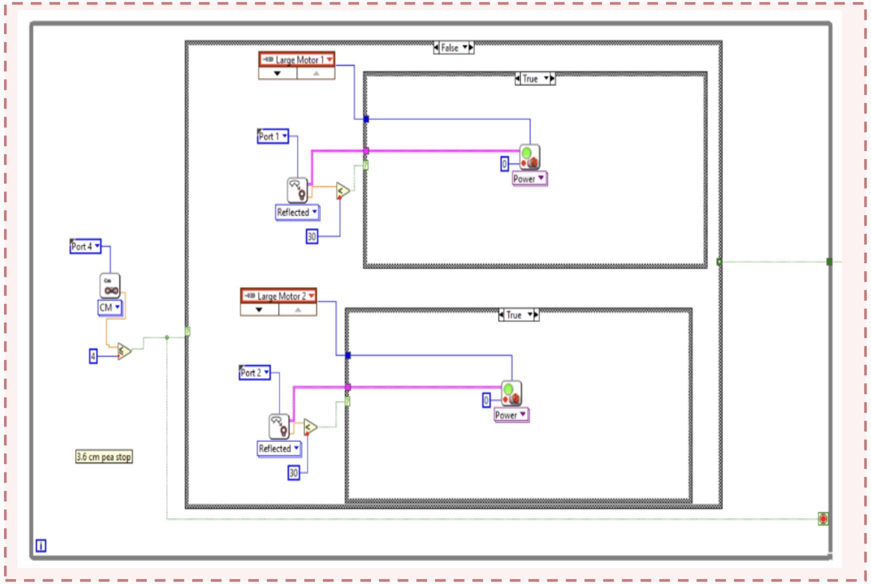
Three Separate LabVIEW Programs Stored on EV3:

1. Line Follower VI Dual-sensor differential steering for closed-loop path

2. Arm Control VI Touch sensor trigger → motor lower/lift sequence

3. Sorting Logic VI Motor load comparison → case structure routes to Station 1 or 2

LabVIEW Code: Line Following



Logic Explanation

1. **Read Sensors**

   Both sensors read reflected light intensity continuously

2. **Sensor A off line**

   Left sensor > threshold of 30 → steer left (motor A slows)

3. **Sensor B off line**

   Right sensor > threshold of 30 → steer right (motor B slows)

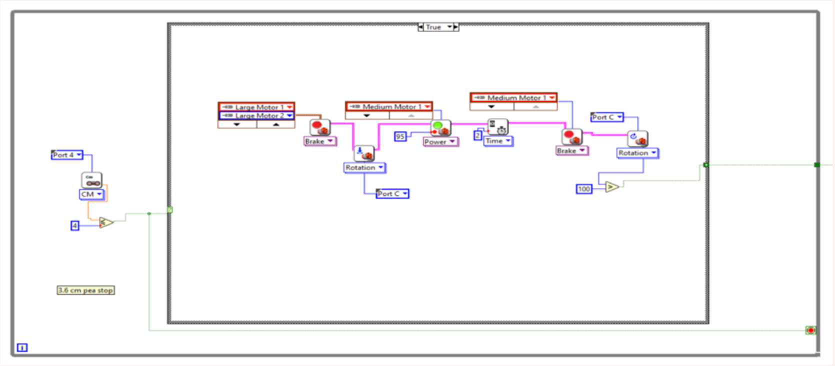
4. **Broken line recovery**

   Maintains heading when line is briefly missing

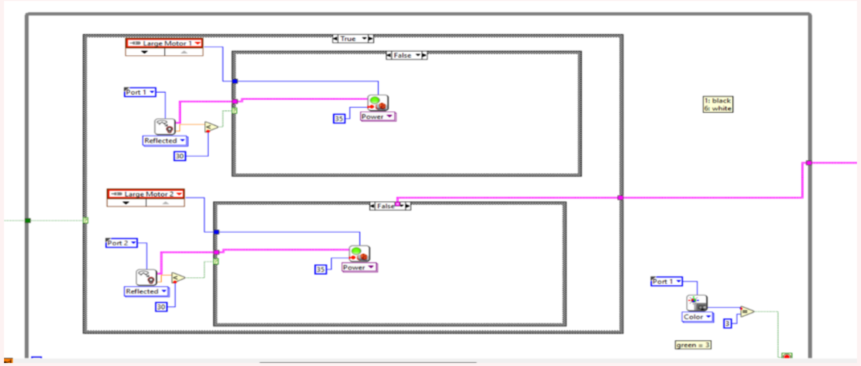
Condition	White Surface	Black Line	Threshold Set
Normal Light	~40–45	~8–12	30
Dim Light	~28–32	~8–10	20 (adjusted)

LabVIEW Code: Bin Detection & Sorting

VI #4 & #5



Arm Control VI

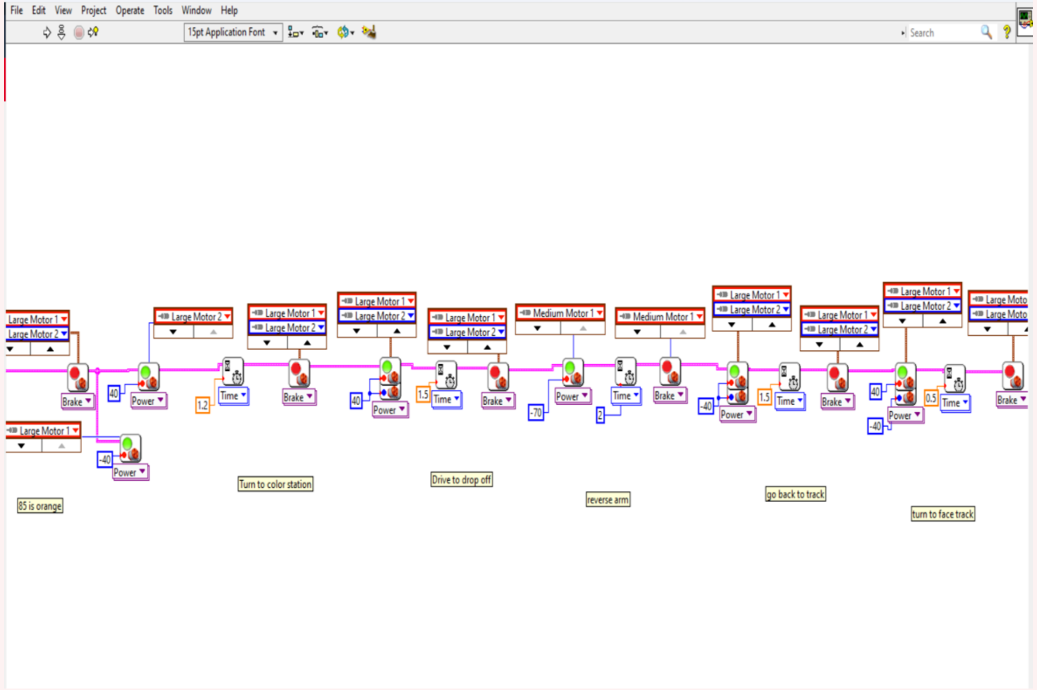


Sorting Logic VI

Ultrasonic sensor detects something in front — Arm Lowers & Grips Bin — Lift → Read Number of rotations — Heavy Load = Metal — Light Load = Organic — Deliver → Return Loop

Sorting Spec: Correctly classify organic vs. metal ≥ 90% of the time | Method: no. of rotations comparison in LabVIEW case structure

LabVIEW Code: Container Exit & Return Loop



Logic Explanation

1 **Container drop-off**

Robot reads the color sensors and according to weight drops off weight at green or red paper

2 **Bin drop-off**

Run medium motor at reverse power
And drop off bin

3 **Join main loop**

Run large motor in negative power
And reverse the robot

4 **Post-delivery return**

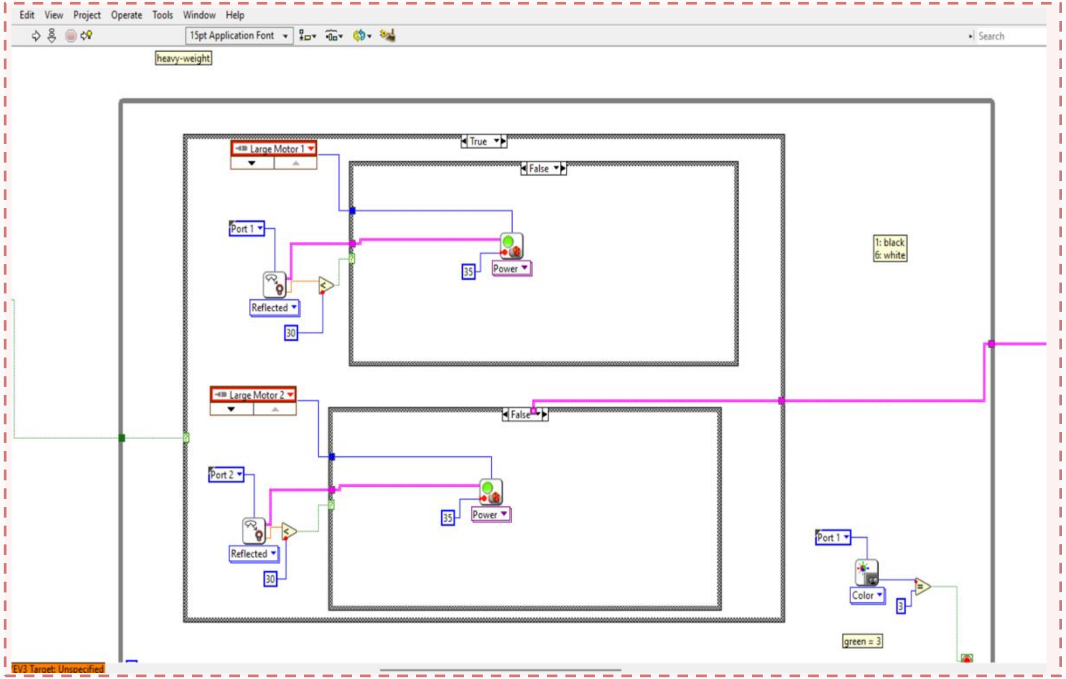
Go back to black line and continue following black line

5 **Cycle repeats**

Continuous until all bins
are sorted and delivered

Spec: Navigate from shipping container to shredder station in ≤ 120 seconds following a black line

LabVIEW Code: Weight Classification Detail



Future Improvement Note:

Analyze weight more effectively instead of just 2 categories

Classification Logic

1. **Arm lifts bin**

   Run medium motor at 95% power for 2 seconds

2. **Load is measured**

   Read number of rotations in 2 seconds

3. **Compare to threshold**

   LabVIEW comparison node checks: no. of rotations > 100?

4. **False → Metal**

   heavy bin → Station 1 (detect green for drop off)

5. **True → Organic**

   light bin → Station 2 (detect red for drop off)

Testing Plan & Results

3 structured tests validated line-following performance against design specs.

Test 1

Lighting Conditions

Spec: Line detection under normal & dim lighting — 10 total runs

10/10 PASS | Normal avg intensity: 19 (SD 0.46) | Dim avg: 12.4 (SD 0.55)

PASS

Test 2

Surface Flatness (Terrain Simulation)

Spec: Line following on flat, slight wrinkle & moderate wrinkle — 15 runs

Flat & slight wrinkle: mostly passed | Moderate wrinkle: 3/5 passed | Avg fail distance: 16.5 in

PARTIAL

Test 3

Offset Re-alignment

Spec: Re-acquire line starting ±0.5 in off-center through a curved section — 10 runs

10/10 PASS | Mean re-acquire time: 1.02s (SD 0.51) | Self-corrected at all offsets

PASS

Demo Performance

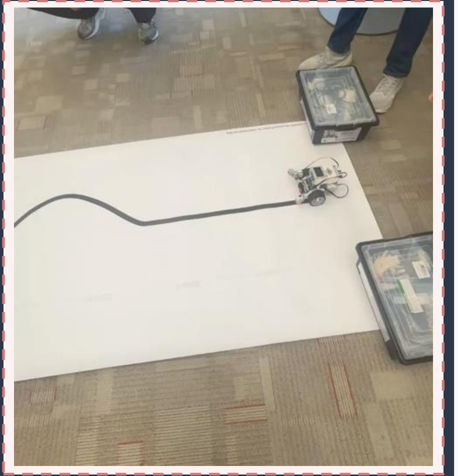
SubTask 1 — Line Following

Zone 1 ✓	Zone 2 ✓	Zone 3 ✓
Zone 4 ✓	Zone 5 ✓	Zone 6 ✓

Station Type: Solid Straight + Solid Curved

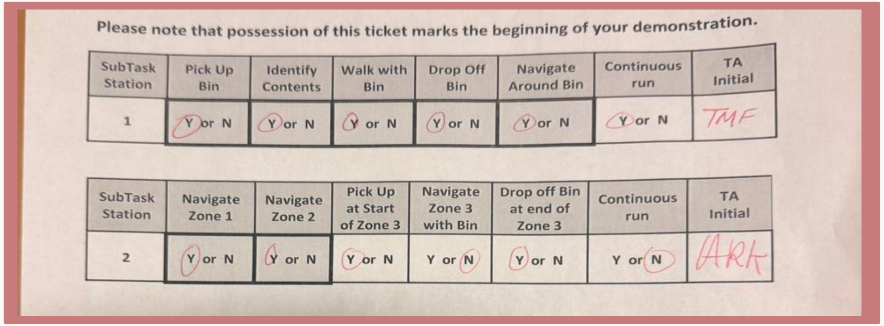
Continuous (no restarts): YES

Time on Task: 47.38 seconds



Final Demo — Full PMR Sequence

- ✓ Follows solid and broken line
- ✓ Arm picks up bin at station
- ✓ Senses weight and drops off at drop off location
- ✓ Exits drop off location autonomously
- ✓ Continues following solid and broken line



Please note that possession of this ticket marks the beginning of your demonstration.

SubTask Station	Pick Up Bin	Identify Contents	Walk with Bin	Drop Off Bin	Navigate Around Bin	Continuous run	TA Initial
1	Y or N	Y or N	Y or N	Y or N	Y or N	Y or N	TMF

SubTask Station	Navigate Zone 1	Navigate Zone 2	Pick Up at Start of Zone 3	Navigate Zone 3 with Bin	Drop off Bin at end of Zone 3	Continuous run	TA Initial
2	Y or N	Y or N	Y or N	Y or N	Y or N	Y or N	ARK

Design Evolution

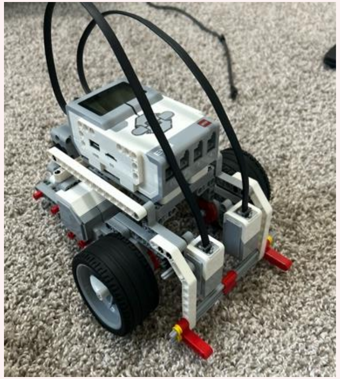
Initial Build

Feb 18–20, 2026

- Basic EV3 build
- Single color sensor (Port 1)
- Focus: SubTask 1 line following
- Issue: EV3 block faulty

Improvements From Status Update 1

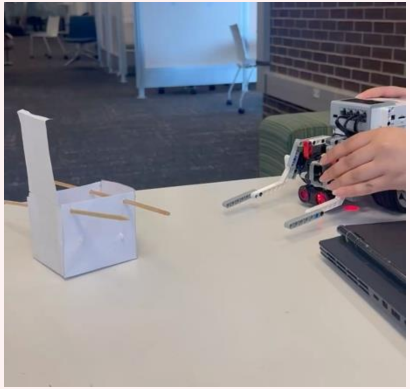
March–April 2026



- Rebuilt the mechanical arm using a slider-crank mechanism
- Removed touch sensors from arm
- Added in medium motor in back

Final Design

March–April 2026



- Rebuilt arm for reliability
- Finalized weight sorting code
- Added container-exit VI
- Completed all testing

Future Enhancements

Strong proof of concept — here's how we'd take it further:

Improved Terrain Handling

Change the design of the color sensors to accommodate a bigger width of the black line.

Dedicated Weight Sensor

Instead of just light or heavy, detect weight accurately.

Obstacle Detection

Readjust according to the obstacle for pick-up.

Thank You

Questions?